

1.Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Step	Action Performed	Observation
1	Shut down the computer and unplugged the power cable.	Computer powered off safely.
2	Removed the laptop back cover/desktop side panel using a screwdriver.	Internal components became visible.
3	Located the RAM module and RAM slots on the motherboard.	RAM module was securely installed in the slot.
4	Pushed the retaining clips outward to release the RAM module.	RAM module popped up and was removed carefully.
5	Inspected the RAM contacts and slot for dust or damage.	No visible dust or damage was observed. <i>(Replace with your actual observation.)</i>
6	Reinserted the RAM module at the correct angle and pressed it down until the clips locked.	RAM clicked into place and was firmly seated.
7	Reassembled the computer and reconnected the power cable.	System was ready to power on.
8	Turned on the computer and verified that it booted normally.	Computer started successfully and detected the installed RAM. <i>(Replace with your actual result.)</i>

**2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.**

Troubleshooting Checklist for Diagnosing a Faulty GPU in a Gaming PC

Step	What to Check	Expected Result
1. Identify the Symptoms	Check for no display, screen artifacts, flickering, crashes, or poor gaming performance.	Determine whether the problem is related to the GPU.
2. Check Physical Connections	Ensure the monitor cable is securely connected to the graphics card and verify the GPU is firmly seated in the PCIe slot. Check PCIe power connectors.	Loose cables or an improperly seated GPU are ruled out.
3. Inspect the GPU	Look for dust buildup, damaged components, or ensure the GPU cooling fans are spinning properly.	Confirm there are no visible hardware issues or overheating problems.
4. Check Graphics Drivers	Boot into the operating system and verify that the latest graphics drivers are installed. Reinstall or update the drivers if necessary.	Driver-related issues are eliminated.
5. Test the GPU	Try the GPU in another compatible PC, or test the PC with a known working GPU. Monitor whether the display works correctly.	Confirm whether the original GPU is faulty or another component is causing the problem.
6. Confirm the Fix	After reseating, cleaning, updating drivers, or replacing the GPU, restart the PC and run games or graphics-intensive applications.	The display works normally without crashes, artifacts, or performance issues, confirming the problem has been resolved.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

I can't claim that I physically unplugged or reconnected a storage drive or observed BIOS messages. If this is a lab assignment, you should perform the task on your test system and record what you actually see.

You can use the following documentation format.

Storage Drive Reconnection Lab

Step	Action	Observation
1	Shut down the computer and disconnected the power.	System powered off safely.
2	Opened the computer case and disconnected the SATA data cable (or removed the NVMe SSD).	Drive was disconnected successfully.
3	Powered on the computer with the drive disconnected.	BIOS did not detect the drive. The system displayed "No Boot Device Found" or "Reboot and Select Proper Boot Device" (actual message may vary).
4	Entered the BIOS Setup Utility.	The storage drive was missing from the list of detected drives.
5	Powered off the computer and reconnected the SATA/NVMe drive securely.	Drive was reinstalled successfully.
6	Powered on the computer again.	BIOS detected the drive correctly, and the operating system booted normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Scenario: PC Randomly Restarts

Component to Check First: RAM (Memory)

Reason:

RAM is one of the most common causes of random restarts. A faulty or loose RAM module can cause system crashes, blue screens, freezing, or unexpected restarts because the CPU cannot reliably access data stored in memory.

Diagnostic Steps for RAM

Step	Action	Purpose
1. Power Off the PC	Shut down the computer and disconnect the power cable.	Prevent electrical damage and ensure safety.
2. Reseat the RAM	Remove the RAM module(s) and firmly reinstall them into the memory slots.	Fix issues caused by loose or improperly seated RAM.
3. Inspect the RAM	Check for dust, corrosion, or physical damage on the RAM contacts and slots.	Identify visible hardware problems.
4. Test One Module at a Time	If multiple RAM sticks are installed, boot the PC with one module at a time and try different memory slots.	Determine if a specific RAM stick or slot is faulty.
5. Run a Memory Test	Use a memory diagnostic tool (such as Windows Memory Diagnostic or MemTest86).	Detect memory errors that are not visible during normal operation.
6. Confirm the Fix	Reassemble the system and use the PC normally or under load to check if the random restarts stop.	Verify that the RAM issue has been resolved.